

Supplementary Material

In this supplementary material, we present additional results related to the proposed NGGD distribution and the NGG regression model. First, plots of the density and survival functions of the NGGD distribution are presented for all quantile levels considered, namely, $\tau \in \{0.10, 0.50, 0.90\}$. These plots illustrate the effect of different parameter settings on the shape of the distribution and the behavior of the survival function.

Next, we present a Monte Carlo simulation study to investigate the finite-sample performance of the parameter estimators for the proposed NGGD distribution and the NGG regression model. For each scenario, 10,000 Monte Carlo replications were considered, with sample sizes $n \in \{100, 200, 400\}$. The performance of the maximum likelihood estimators was evaluated in terms of mean, bias, standard deviation (SD), and mean squared error (MSE). In addition, confidence intervals constructed at the 95% confidence level were evaluated using coverage probability (CP).

The Monte Carlo study was implemented in R using computational routines developed by the authors. Parameter estimation was carried out by maximum likelihood, with numerical optimization based on the Nelder–Mead algorithm. To reduce numerical instability and facilitate convergence of the iterative procedure, a preliminary step was adopted to obtain initial values.

In this step, preliminary estimates were obtained by minimizing the sum of squares of the score vector. This procedure used the `nls.lm()` function from the `minpack.lm` package, which employs a modification of the Levenberg–Marquardt algorithm to minimize the sum of squares of the vector returned by the objective function. The vector of initial values was defined as follows: for the parameter μ_τ , the empirical quantile corresponding to level τ was used; for the parameters σ and ν , random initial values were generated from a uniform distribution on the interval $(0, 2)$.

Using these preliminary values, the final parameter estimates were obtained with the Nelder–Mead algorithm. During the simulation study, the nonconvergence rate ranged from 10% to 15% for the NGGD distribution, whereas it was approximately 10% for the NGGD regression model.

1. Graphical behavior of the NGGD distribution – Density function

This section presents plots of the density function of the $\text{NGGD}(\mu_\tau, \sigma, \nu)$ distribution for different settings of the parameters μ_τ , σ , and ν , considering the quantile levels $\tau \in \{0.10, 0.50, 0.90\}$. These plots show how changes in the parameters affect the location, concentration, and shape of the density over the positive support.

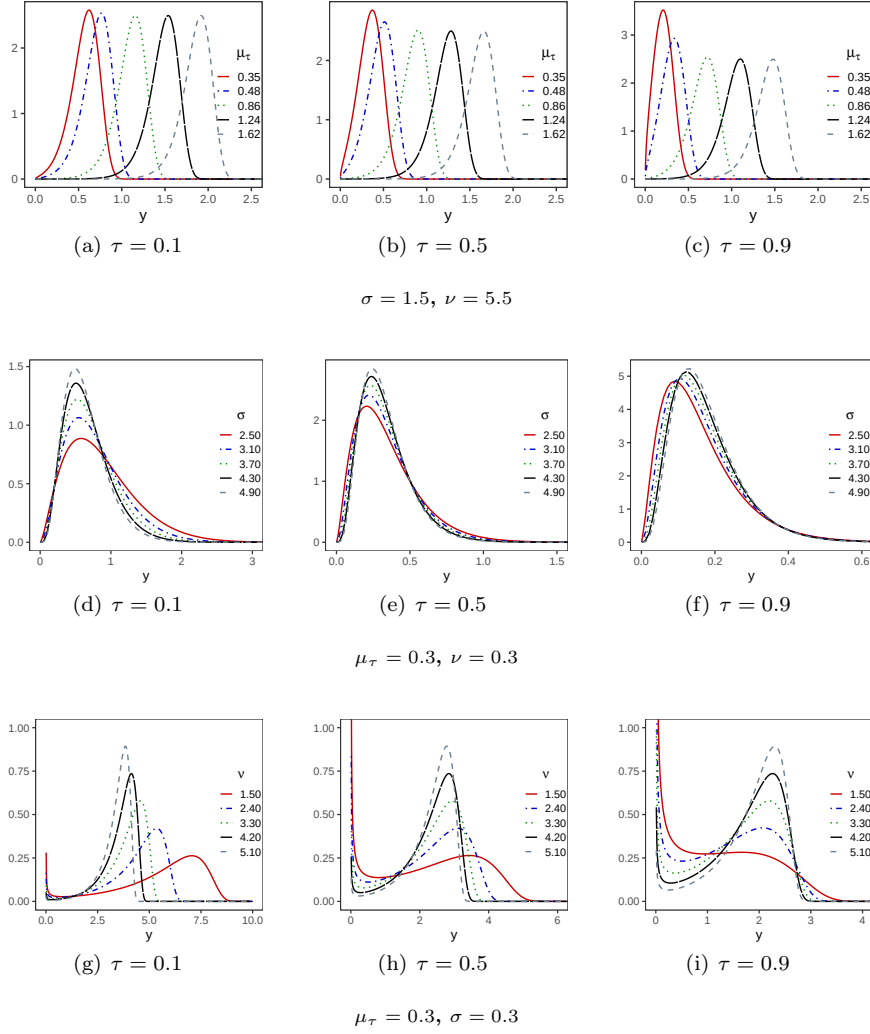


FIGURE 1: Probability density functions of the $\text{NGGD}(\mu_\tau, \sigma, \nu)$ distribution for $\tau = \{0.1, 0.5, 0.9\}$, considering variations in μ_τ , σ , and ν .

2. Graphical behavior of the NGGD distribution – Survival function

The survival function of the $\text{NGGD}(\mu_\tau, \sigma, \nu)$ distribution is examined for the quantile levels $\tau \in \{0.10, 0.50, 0.90\}$. The plots presented in this section describe the probability that the random variable exceeds y under different combinations of the parameters μ_τ , σ , and ν . Thus, they show how these combinations affect the decay pattern of the survival function over the positive support.

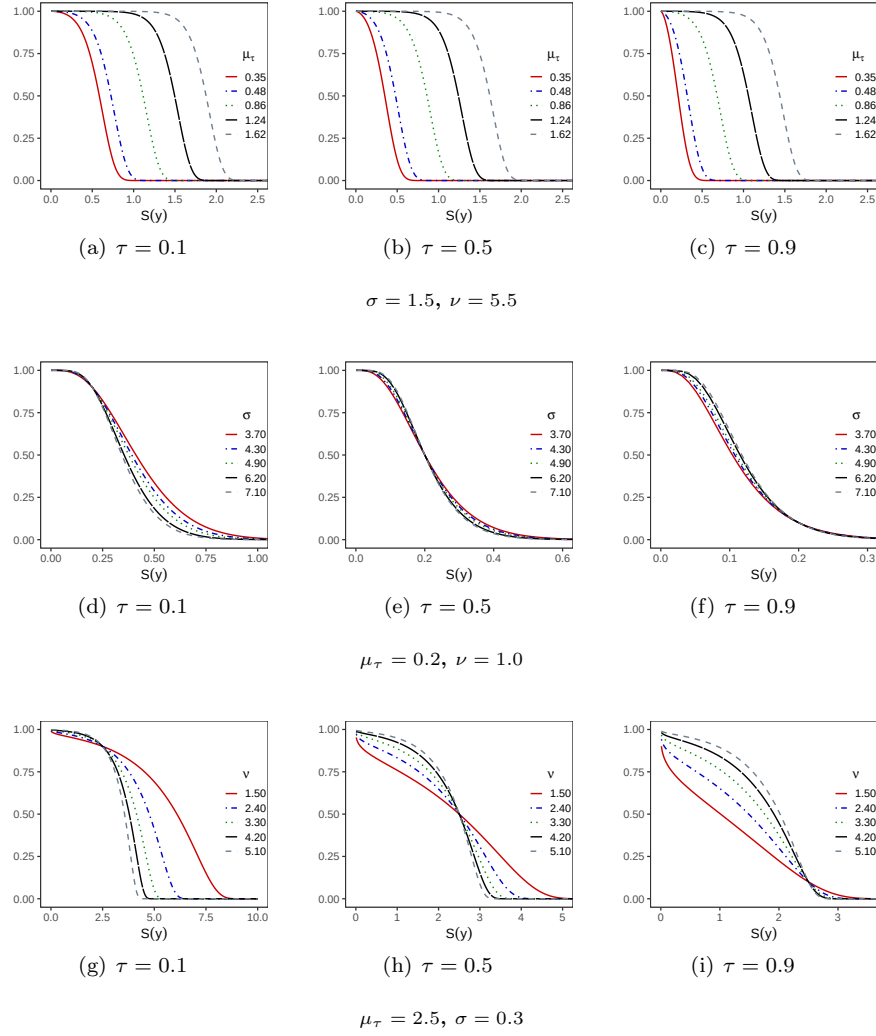


FIGURE 2: Survival functions of the $\text{NGGD}(\mu_\tau, \sigma, \nu)$ distribution for $\tau = (0.1, 0.5, 0.9)$, considering variations in μ_τ , σ , and ν .

3. Monte Carlo simulations for the NGGD distribution

This section presents the results of the Monte Carlo simulations conducted to evaluate the finite-sample performance of the maximum likelihood estimators of the parameters of the NGGD distribution.

3.1. Scenario 1 – $\mu_\tau = 2.85$, $\sigma = 1.5$, and $\nu = 0.5$

TABLE 1: Monte Carlo simulation results for the NGGD(μ_τ, σ, ν) distribution under one parametric scenario, with $\mu_\tau = 2.85$, $\sigma = 1.5$, and $\nu = 0.5$. Results are reported for sample sizes $n \in \{100, 200, 400, 800\}$ and quantile parameters $\tau \in \{0.10, 0.50, 0.90\}$.

Measures	n	$\tau = 0.10$			$\tau = 0.50$			$\tau = 0.90$		
		$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$
Mean	100	2.8715	1.6117	0.5396	2.8462	1.5486	0.5263	2.8360	1.5392	0.5169
	200	2.8649	1.5408	0.5217	2.8484	1.5224	0.5118	2.8444	1.5205	0.5083
	400	2.8559	1.5219	0.5085	2.8491	1.5105	0.5063	2.8469	1.5079	0.5043
	800	2.8519	1.5123	0.5033	2.8496	1.5041	0.5036	2.8479	1.5037	0.5023
Bias	100	0.0215	0.1117	0.0396	-0.0038	0.0486	0.0263	-0.0140	0.0392	0.0169
	200	0.0149	0.0408	0.0217	-0.0016	0.0224	0.0118	-0.0056	0.0205	0.0083
	400	0.0059	0.0219	0.0085	-0.0009	0.0105	0.0063	-0.0031	0.0079	0.0043
	800	0.0019	0.0123	0.0033	-0.0004	0.0041	0.0036	-0.0021	0.0037	0.0023
SD	100	0.3152	0.4951	0.2289	0.1294	0.3434	0.1521	0.0922	0.3009	0.1162
	200	0.2249	0.3043	0.1427	0.0918	0.2261	0.0989	0.0649	0.2064	0.0781
	400	0.1598	0.1989	0.0938	0.0657	0.1548	0.0664	0.0460	0.1436	0.0538
	800	0.1137	0.1376	0.0640	0.0463	0.1087	0.0493	0.0324	0.0995	0.0375
MSE	100	0.0998	0.2576	0.0539	0.0168	0.1203	0.0238	0.0087	0.0921	0.0138
	200	0.0508	0.0943	0.0208	0.0084	0.0516	0.0099	0.0042	0.0430	0.0062
	400	0.0256	0.0400	0.0089	0.0043	0.0241	0.0045	0.0021	0.0207	0.0029
	800	0.0129	0.0191	0.0041	0.0021	0.0118	0.0024	0.0011	0.0099	0.0014
CP	100	0.8934	0.8708	0.8987	0.9387	0.9248	0.9445	0.9383	0.9493	0.9511
	200	0.9061	0.9034	0.9247	0.9430	0.9339	0.9479	0.9421	0.9457	0.9491
	400	0.9149	0.9203	0.9305	0.9398	0.9375	0.9437	0.9467	0.9436	0.9450
	800	0.9140	0.9245	0.9304	0.9459	0.9312	0.9393	0.9478	0.9406	0.9441

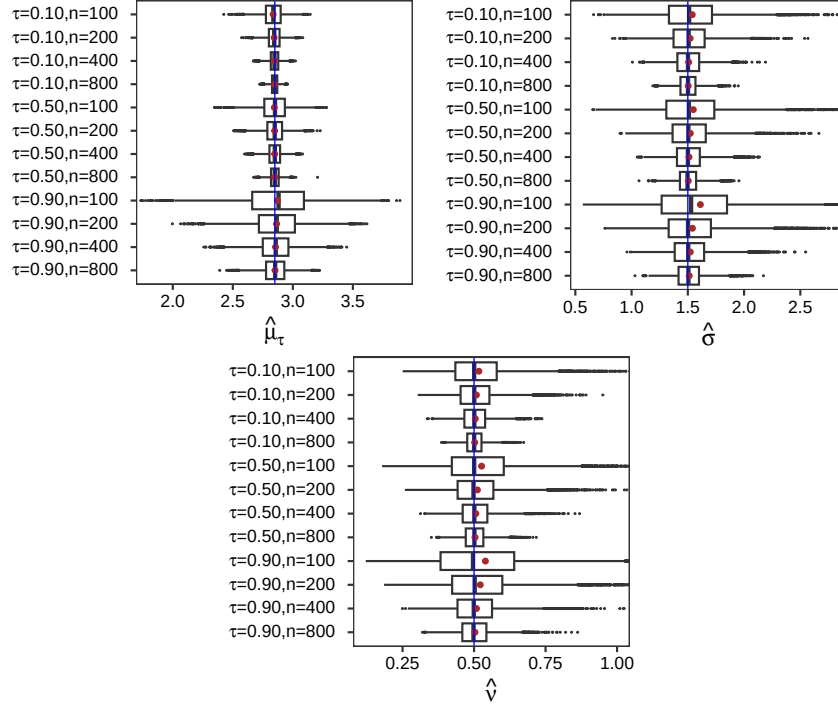


FIGURE 3: Boxplots of the maximum likelihood estimates obtained from Monte Carlo simulations for the $\text{NGGD}(\mu_\tau, \sigma, \nu)$ distribution, with $\mu_\tau = 2.85$, $\sigma = 1.5$, and $\nu = 0.5$, considering $n \in \{100, 200, 400, 800\}$ and $\tau \in \{0.10, 0.50, 0.90\}$.
Legend: —: true parameter value; •: mean of the 10,000 replications.

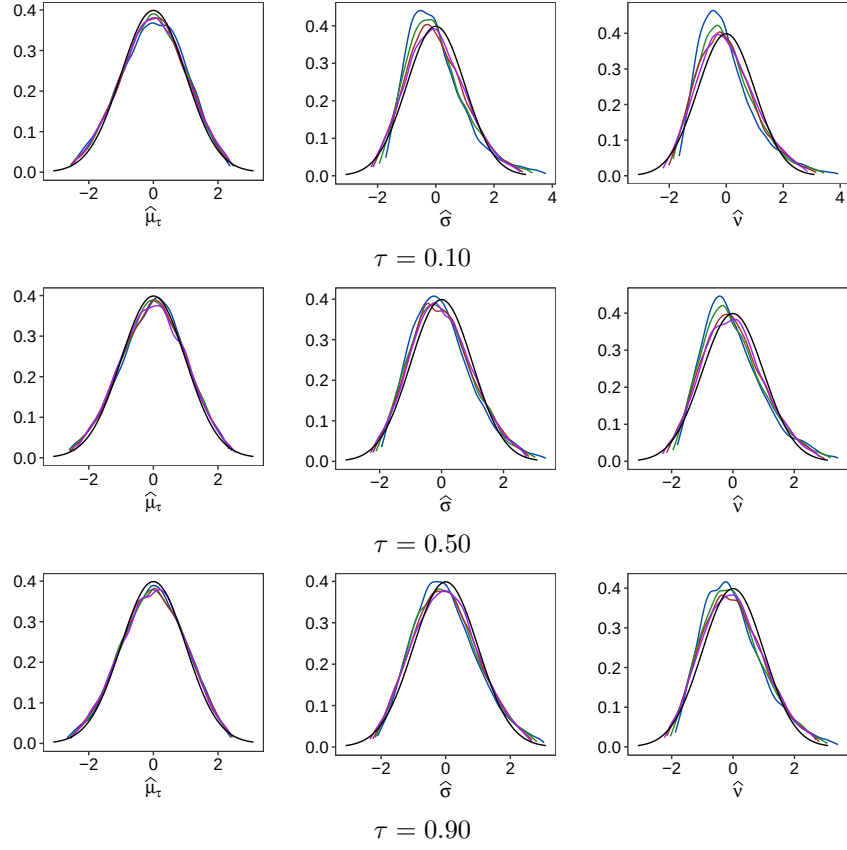
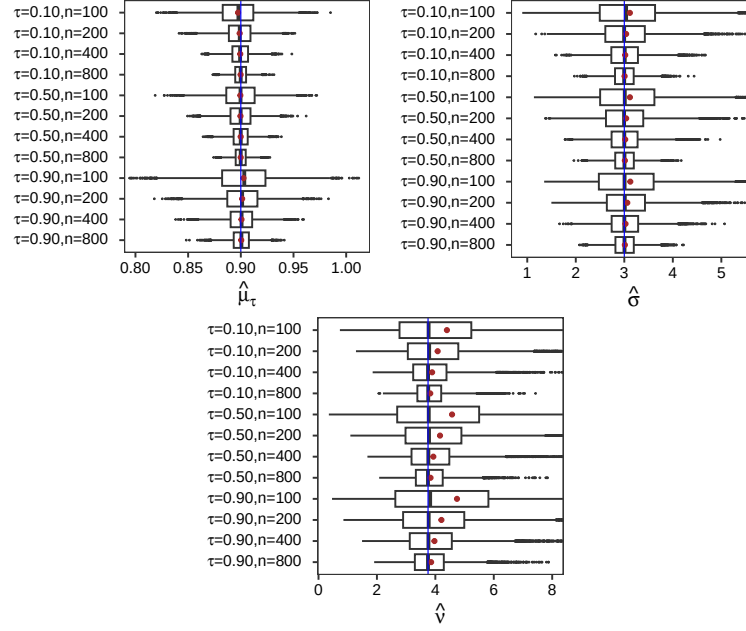


FIGURE 4: Sampling densities of the maximum likelihood estimators of the parameters of the $\text{NGGD}(\mu_\tau, \sigma, \nu)$ distribution obtained via Monte Carlo simulation, considering $\mu_\tau = 2.85$, $\sigma = 1.5$, and $\nu = 0.5$, $\tau \in \{0.10, 0.50, 0.90\}$ and $n \in \{100, 200, 400, 800\}$. Legend: $n = 100$; $n = 200$; $n = 400$; $n = 800$; $N(0,1)$.

3.2. Scenario 2 – $\mu_\tau = 0.9$, $\sigma = 3.0$, and $\nu = 3.75$

TABLE 2: Monte Carlo simulation results for the NGGD(μ_τ, σ, ν) distribution under one parametric scenario, with $\mu_\tau = 0.9$, $\sigma = 3$, and $\nu = 3.75$. Results are reported for sample sizes $n \in \{100, 200, 400, 800\}$ and quantile parameters $\tau \in \{0.10, 0.50, 0.90\}$.

Measures	n	$\tau = 0.10$			$\tau = 0.50$			$\tau = 0.90$		
		$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$
Mean	100	0.9026	3.1241	4.7409	0.8996	3.1161	4.5718	0.8974	3.1150	4.3958
	200	0.9013	3.0617	4.2079	0.8997	3.0388	4.1598	0.8989	3.0381	4.0796
	400	0.9006	3.0252	3.9689	0.8999	3.0189	3.9312	0.8995	3.0193	3.8895
	800	0.9004	3.0118	3.8555	0.8999	3.0118	3.8352	0.8998	3.0068	3.8258
Bias	100	0.0026	0.1241	0.9909	-0.0004	0.1161	0.8218	-0.0026	0.1150	0.6458
	200	0.0013	0.0617	0.4579	-0.0003	0.0388	0.4098	-0.0011	0.0381	0.3296
	400	0.0006	0.0252	0.2189	-0.0001	0.0189	0.1812	-0.0005	0.0193	0.1395
	800	0.0004	0.0118	0.1055	-0.0001	0.0118	0.0852	-0.0002	0.0068	0.0758
SD	100	0.0305	0.9265	3.0915	0.0200	0.9016	2.7942	0.0217	0.8954	2.4119
	200	0.0214	0.6069	1.9287	0.0141	0.5980	1.7409	0.0154	0.6051	1.4873
	400	0.0155	0.4115	1.2122	0.0098	0.4053	1.0495	0.0109	0.4134	0.9264
	800	0.0108	0.2830	0.7892	0.0071	0.2856	0.7128	0.0077	0.2905	0.6310
MSE	100	0.0009	0.8738	10.5396	0.0004	0.8263	8.4831	0.0005	0.8150	6.2344
	200	0.0005	0.3722	3.9296	0.0002	0.3592	3.1986	0.0002	0.3676	2.3207
	400	0.0002	0.1700	1.5174	0.0001	0.1647	1.1342	0.0001	0.1713	0.8778
	800	0.0001	0.0802	0.6340	0.0000	0.0817	0.5153	0.0001	0.0845	0.4038
CP	100	0.9310	0.9580	0.9007	0.9417	0.9602	0.9144	0.9328	0.9635	0.9217
	200	0.9392	0.9529	0.9254	0.9440	0.9531	0.9308	0.9405	0.9533	0.9367
	400	0.9385	0.9495	0.9378	0.9516	0.9521	0.9476	0.9434	0.9512	0.9499
	800	0.9414	0.9515	0.9479	0.9493	0.9487	0.9465	0.9485	0.9502	0.9482



(b) Scenario 2: $\mu_\tau = 1.86$, $\sigma = 0.93$ and $\nu = 0.42$

FIGURE 5: Boxplots of the maximum likelihood estimates obtained from Monte Carlo simulations for the $\text{NGGD}(\mu_\tau, \sigma, \nu)$ distribution, with $\mu_\tau = 0.9$, $\sigma = 3.0$, and $\nu = 3.75$, considering $n \in \{100, 200, 400, 800\}$ and $\tau \in \{0.10, 0.50, 0.90\}$. Legend: —: true parameter value; •: mean of the 10,000 replications.

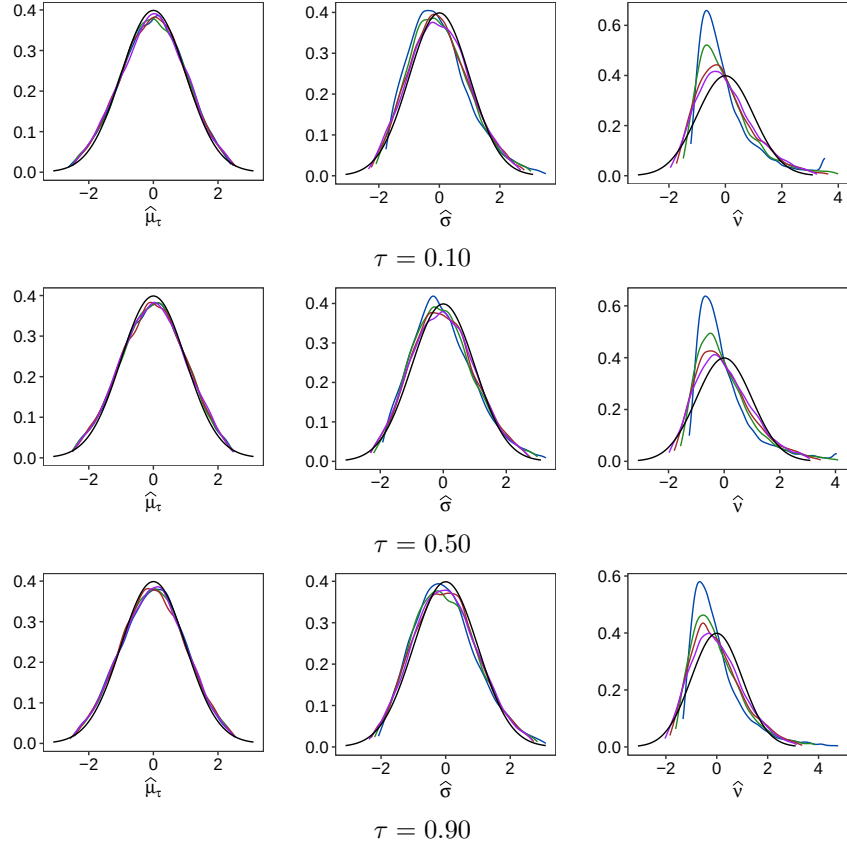


FIGURE 6: Sampling densities of the maximum likelihood estimators of the parameters of the $\text{NGGD}(\mu_\tau, \sigma, \nu)$ distribution obtained via Monte Carlo simulation, considering $\mu_\tau = 0.9$, $\sigma = 3.0$, and $\nu = 3.75$, $\tau \in \{0.1, 0.5, 0.9\}$ and $n \in \{100, 200, 400, 800\}$. Legend: $n = 100$; $n = 200$; $n = 400$; $n = 800$; $N(0,1)$.

3.3. Scenario 3 – $\mu_\tau = 3.65$, $\sigma = 2.4$, and $\nu = 2.0$

TABLE 3: Monte Carlo simulation results for the NGGD(μ_τ, σ, ν) distribution under one parametric scenario, with $\mu_\tau = 3.65$, $\sigma = 2.4$, and $\nu = 2$. Results are reported for sample sizes $n \in \{100, 200, 400, 800\}$ and quantile parameters $\tau \in \{0.10, 0.50, 0.90\}$.

Measures	n	$\tau = 0.10$			$\tau = 0.50$			$\tau = 0.90$		
		$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$
Mean	100	3.6523	2.5659	2.5595	3.6502	2.5604	2.5477	3.6463	2.5572	2.5345
	200	3.6514	2.4806	2.2089	3.6502	2.4746	2.2244	3.6485	2.4599	2.2440
	400	3.6510	2.4323	2.1004	3.6499	2.4319	2.1036	3.6490	2.4354	2.0898
	800	3.6508	2.4166	2.0487	3.6500	2.4155	2.0480	3.6493	2.4163	2.0454
Bias	100	0.0023	0.1659	0.5595	0.0002	0.1604	0.5477	-0.0037	0.1572	0.5345
	200	0.0014	0.0806	0.2089	0.0002	0.0746	0.2244	-0.0015	0.0599	0.2440
	400	0.0010	0.0323	0.1004	-0.0001	0.0319	0.1036	-0.0010	0.0354	0.0898
	800	0.0008	0.0166	0.0487	0.0000	0.0155	0.0480	-0.0007	0.0163	0.0454
SD	100	0.0611	0.8111	2.0650	0.0360	0.8004	2.0062	0.0345	0.7936	1.9845
	200	0.0436	0.5104	1.0286	0.0251	0.5041	1.0637	0.0246	0.4987	1.0517
	400	0.0310	0.3342	0.6140	0.0179	0.3368	0.6228	0.0174	0.3316	0.5961
	800	0.0217	0.2319	0.4042	0.0126	0.2342	0.4058	0.0122	0.2297	0.3976
MSE	100	0.0037	0.6854	4.5773	0.0013	0.6664	4.3246	0.0012	0.6545	4.2240
	200	0.0019	0.2670	1.1016	0.0006	0.2597	1.1817	0.0006	0.2523	1.1656
	400	0.0010	0.1127	0.3871	0.0003	0.1145	0.3986	0.0003	0.1112	0.3634
	800	0.0005	0.0541	0.1657	0.0002	0.0551	0.1670	0.0001	0.0530	0.1601
CP	100	0.9351	0.9482	0.8890	0.9404	0.9521	0.8887	0.9363	0.9508	0.8891
	200	0.9385	0.9482	0.9143	0.9441	0.9451	0.9183	0.9441	0.9452	0.9218
	400	0.9408	0.9516	0.9382	0.9454	0.9497	0.9327	0.9436	0.9543	0.9357
	800	0.9424	0.9502	0.9434	0.9465	0.9490	0.9403	0.9496	0.9526	0.9447

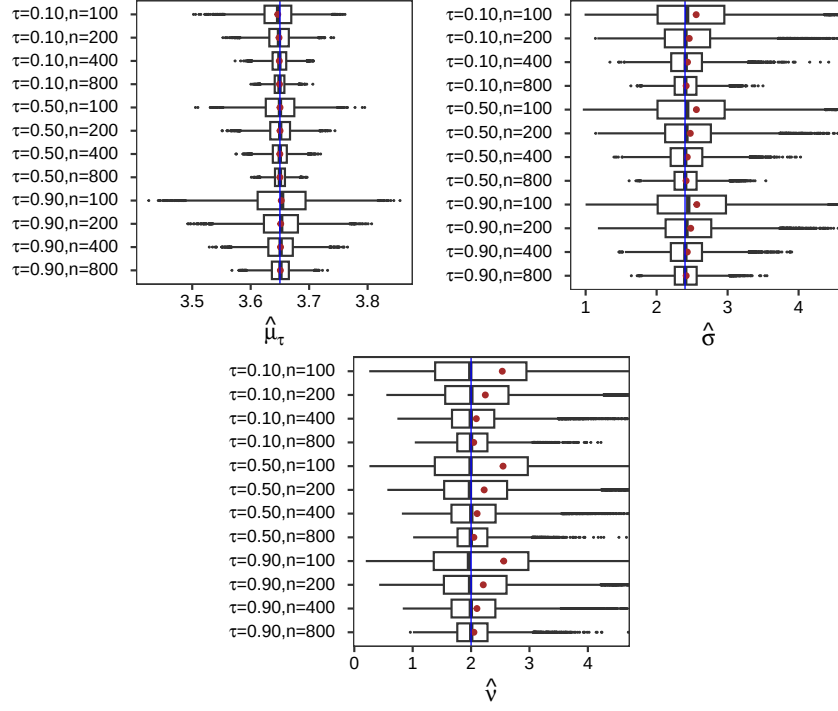


FIGURE 7: Boxplots of the maximum likelihood estimates obtained from Monte Carlo simulations for the $\text{NGGD}(\mu_\tau, \sigma, \nu)$ distribution, with $\mu_\tau = 3.65$, $\sigma = 2.4$, and $\nu = 2.0$, considering $n \in \{100, 200, 400\}$ and $\tau \in \{0.10, 0.50, 0.90\}$. Legend: --- : true parameter value; $\bullet$: mean of the 10,000 replications.

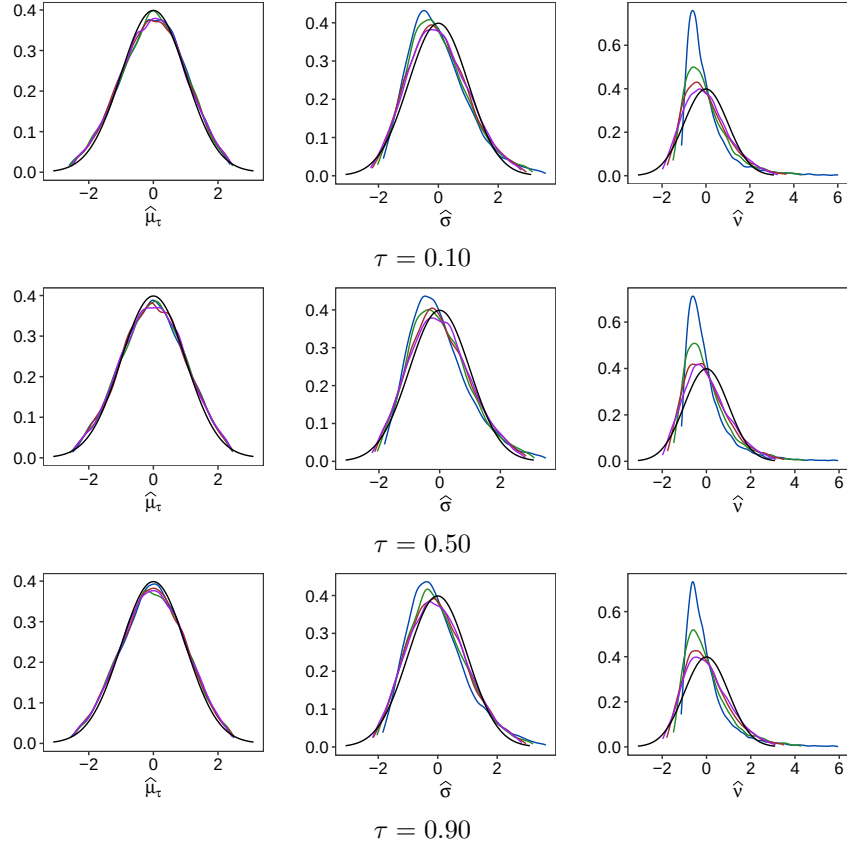


FIGURE 8: Sampling densities of the maximum likelihood estimators of the parameters of the $\text{NGGD}(\mu_\tau, \sigma, \nu)$ distribution obtained via Monte Carlo simulation, considering $\mu_\tau = 3.65$, $\sigma = 2.4$, and $\nu = 2.0$, $\tau \in \{0.1, 0.5, 0.9\}$ and $n \in \{100, 200, 400, 800\}$. Legend: $n = 100$; $n = 200$; $n = 400$; $n = 800$; $N(0,1)$.

3.4. Scenario 4 – $\mu_\tau = 1.85$, $\sigma = 0.95$, and $\nu = 0.5$

TABLE 4: Monte Carlo simulation results for the NGGD(μ_τ, σ, ν) distribution under one parametric scenario, with $\mu_\tau = 1.85$, $\sigma = 0.95$, and $\nu = 0.5$. Results are reported for sample sizes $n \in \{100, 200, 400, 800\}$ and quantile parameters $\tau \in \{0.10, 0.50, 0.90\}$.

Measures	n	$\tau = 0.10$			$\tau = 0.50$			$\tau = 0.90$		
		$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$
Mean	100	1.9036	0.9913	0.5285	1.8607	0.9814	0.5124	1.8326	0.9993	0.5079
	200	1.8762	0.9687	0.5126	1.8506	0.9610	0.5068	1.8413	0.9762	0.5036
	400	1.8648	0.9552	0.5085	1.8510	0.9567	0.5033	1.8460	0.9626	0.5016
	800	1.8561	0.9535	0.5035	1.8516	0.9534	0.5013	1.8472	0.9555	0.5010
Bias	100	0.0536	0.0413	0.0285	0.0107	0.0314	0.0124	-0.0174	0.0493	0.0079
	200	0.0262	0.0187	0.0126	0.0006	0.0110	0.0068	-0.0087	0.0262	0.0036
	400	0.0148	0.0052	0.0085	0.0010	0.0067	0.0033	-0.0040	0.0126	0.0016
	800	0.0061	0.0035	0.0035	0.0016	0.0034	0.0013	-0.0028	0.0055	0.0010
SD	100	0.4152	0.2418	0.1717	0.1741	0.1960	0.0997	0.1204	0.2606	0.0836
	200	0.2967	0.1529	0.1068	0.1244	0.1323	0.0683	0.0856	0.1840	0.0575
	400	0.2098	0.1036	0.0737	0.0878	0.0933	0.0482	0.0605	0.1237	0.0400
	800	0.1500	0.0730	0.0505	0.1452	0.0655	0.0334	0.0425	0.0873	0.0282
MSE	100	0.1753	0.0602	0.0303	0.0304	0.0394	0.0101	0.0148	0.0703	0.0070
	200	0.0887	0.0237	0.0116	0.0155	0.0176	0.0047	0.0074	0.0346	0.0033
	400	0.0442	0.0108	0.0055	0.0077	0.0087	0.0023	0.0037	0.0155	0.0016
	800	0.0225	0.0053	0.0026	0.0211	0.0043	0.0011	0.0018	0.0077	0.0008
CP	100	0.9031	0.9188	0.9350	0.9389	0.9560	0.9530	0.9295	0.9548	0.9487
	200	0.9153	0.9363	0.9418	0.9406	0.9512	0.9518	0.9407	0.9444	0.9447
	400	0.9207	0.9371	0.9427	0.9418	0.9461	0.9444	0.9437	0.9457	0.9434
	800	0.9204	0.9343	0.9369	0.9453	0.9424	0.9469	0.9491	0.9442	0.9444

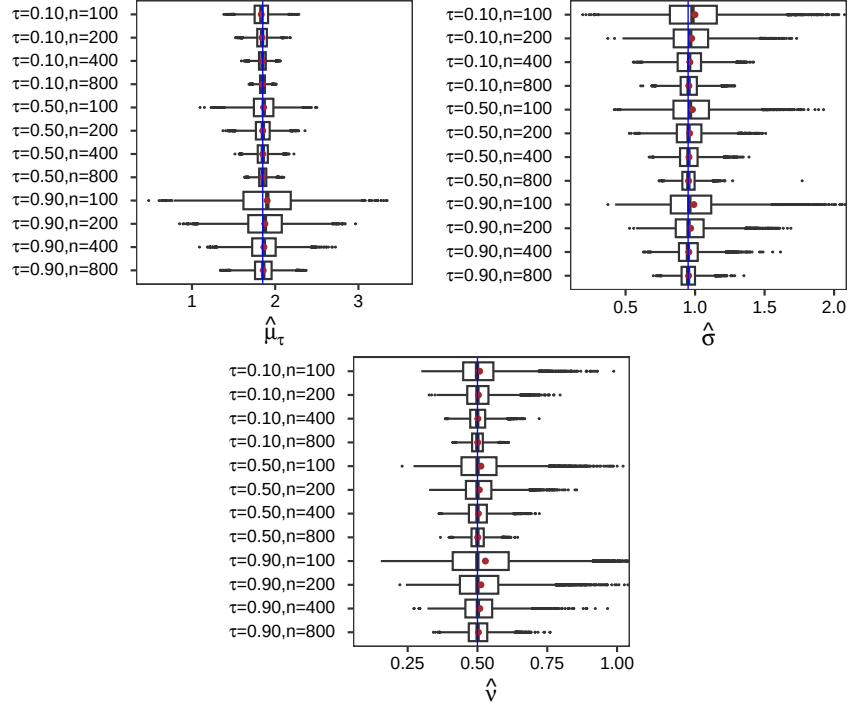


FIGURE 9: Boxplots of the maximum likelihood estimates obtained from Monte Carlo simulations for the $\text{NGGD}(\mu_\tau, \sigma, \nu)$ distribution, with $\mu_\tau = 1.85$, $\sigma = 0.95$, and $\nu = 0.5$, considering $n \in \{100, 200, 400\}$ and $\tau \in \{0.10, 0.50, 0.90\}$. Legend: $\bullet$: true parameter value; $\bullet$: mean of the 10,000 replications.

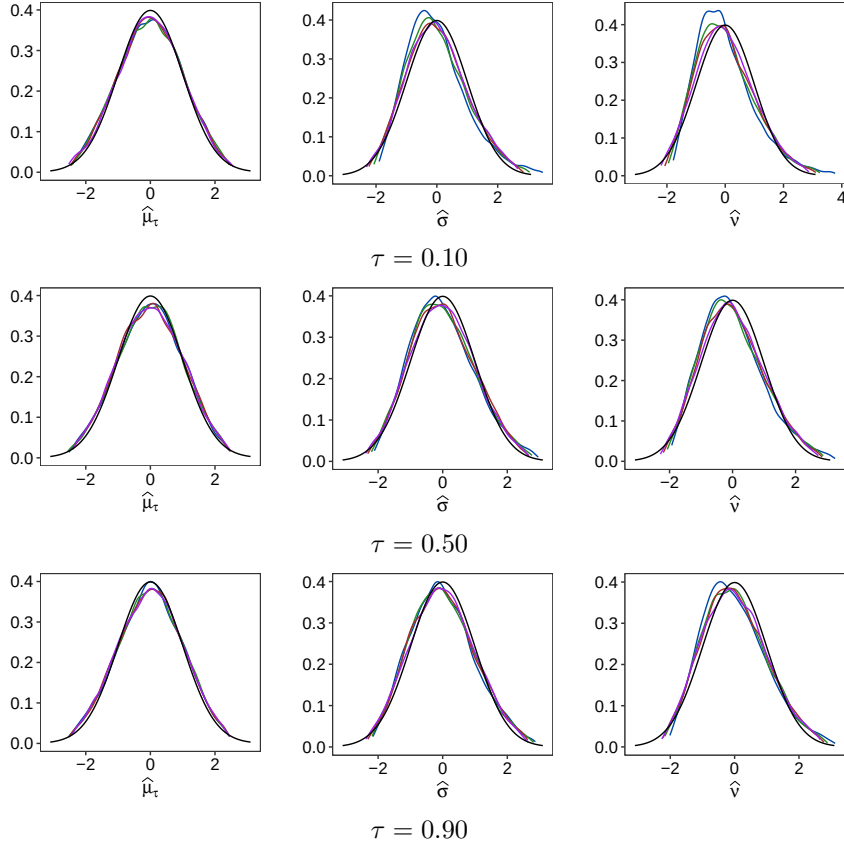


FIGURE 10: Sampling densities of the maximum likelihood estimators of the parameters of the $\text{NGGD}(\mu_\tau, \sigma, \nu)$ distribution obtained via Monte Carlo simulation, considering $\mu_\tau = 1.85$, $\sigma = 0.95$, and $\nu = 0.5$, $\tau \in \{0.1, 0.5, 0.9\}$ and $n \in \{100, 200, 400, 800\}$. Legend: $n = 100$; $n = 200$; $n = 400$; $n = 800$; $N(0,1)$.

4. NGG regression

This section presents additional results from the Monte Carlo simulations for the NGG regression model. The tables report the mean estimates, biases, standard deviations, mean squared errors, and coverage probabilities of the 95% confidence intervals for the scenarios, sample sizes, and quantiles considered.

The plots complement the analysis with boxplots and sampling densities of the maximum likelihood estimators, showing the dispersion of the estimates, their concentration around the true values, and the asymptotic behavior of the estimators.

4.1. Scenario 1 – $\sigma = 1.5$, $\nu = 0.5$, $\beta = (0.5, 0.3, 0.2)$

TABLE 5: Monte Carlo simulation results for the maximum likelihood estimators of the NGG regression model with $\sigma = 1.5$, $\nu = 0.5$, $\beta_0 = 0.5$, $\beta_1 = 0.3$, $\beta_2 = 0.2$, $n \in \{100, 200, 400, 800\}$, and $\tau \in \{0.10, 0.50, 0.90\}$.

par	τ	$n = 100$					$n = 200$				
		mean	bias	se	mse	cp	mean	bias	se	mse	cp
$\hat{\sigma}$	0.1	1.5960	0.0960	0.5763	0.3414	0.9356	1.5419	0.0419	0.3586	0.1304	0.9457
	0.5	1.5580	0.0580	0.4354	0.1930	0.9410	1.5319	0.0319	0.2827	0.0809	0.9483
	0.9	1.5422	0.0422	0.3697	0.1385	0.9472	1.5210	0.0210	0.2432	0.0596	0.9525
$\hat{\nu}$	0.1	0.5300	0.0300	0.1323	0.0184	0.9636	0.5142	0.0142	0.0851	0.0074	0.9567
	0.5	0.5397	0.0397	0.1516	0.0246	0.9603	0.5172	0.0172	0.1015	0.0106	0.9505
	0.9	0.5595	0.0595	0.2137	0.0492	0.9524	0.5302	0.0302	0.1426	0.0212	0.9526
$\hat{\beta}_0$	0.1	0.5142	0.0142	0.1806	0.0328	0.9356	0.5033	0.0033	0.1306	0.0171	0.9421
	0.5	0.5042	0.0042	0.1319	0.0174	0.9428	0.4997	-0.0003	0.0990	0.0098	0.9479
	0.9	0.4835	-0.0165	0.1525	0.0235	0.9396	0.4926	-0.0074	0.1033	0.0107	0.9449
$\hat{\beta}_1$	0.1	0.2940	-0.0060	0.1879	0.0353	0.9482	0.3019	0.0019	0.1289	0.0166	0.9496
	0.5	0.2986	-0.0014	0.1777	0.0316	0.9425	0.2994	-0.0006	0.1292	0.0167	0.9460
	0.9	0.2973	-0.0027	0.1938	0.0376	0.9412	0.2982	-0.0018	0.1238	0.0153	0.9461
$\hat{\beta}_2$	0.1	0.1973	-0.0027	0.1827	0.0334	0.9442	0.2012	0.0012	0.1213	0.0147	0.9504
	0.5	0.1941	-0.0059	0.1741	0.0303	0.9424	0.2011	0.0011	0.1265	0.0160	0.9466
	0.9	0.2024	0.0024	0.1788	0.0320	0.9405	0.1987	-0.0013	0.1345	0.0181	0.9426
par	τ	$n = 400$					$n = 800$				
		mean	bias	se	mse	cp	mean	bias	se	mse	cp
$\hat{\sigma}$	0.1	1.5208	0.0208	0.2430	0.0595	0.9448	1.5122	0.0122	0.1646	0.0272	0.9492
	0.5	1.5173	0.0173	0.1964	0.0389	0.9456	1.5059	0.0059	0.1303	0.0170	0.9515
	0.9	1.5093	0.0093	0.1673	0.0281	0.9474	1.5049	0.0049	0.1163	0.0135	0.9522
$\hat{\nu}$	0.1	0.5066	0.0066	0.0583	0.0034	0.9538	0.5031	0.0031	0.0403	0.0016	0.9556
	0.5	0.5081	0.0081	0.0702	0.0050	0.9473	0.5044	0.0044	0.0474	0.0023	0.9536
	0.9	0.5156	0.0156	0.0981	0.0099	0.9504	0.5070	0.0070	0.0672	0.0046	0.9529
$\hat{\beta}_0$	0.1	0.5012	0.0012	0.0909	0.0083	0.9498	0.5003	0.0003	0.0648	0.0042	0.9515
	0.5	0.5007	0.0007	0.0703	0.0049	0.9460	0.5000	0.0000	0.0510	0.0026	0.9523
	0.9	0.4955	-0.0045	0.0701	0.0049	0.9499	0.4988	-0.0012	0.0501	0.0025	0.9493
$\hat{\beta}_1$	0.1	0.3008	0.0008	0.0910	0.0083	0.9548	0.3003	0.0003	0.0639	0.0041	0.9488
	0.5	0.2994	-0.0006	0.0899	0.0081	0.9505	0.3001	0.0001	0.0621	0.0039	0.9507
	0.9	0.2994	-0.0006	0.0884	0.0078	0.9508	0.2987	-0.0013	0.0632	0.0040	0.9522
$\hat{\beta}_2$	0.1	0.2008	0.0008	0.0889	0.0079	0.9515	0.2022	0.0022	0.0643	0.0041	0.9530
	0.5	0.1996	-0.0004	0.0895	0.0080	0.9514	0.1994	-0.0006	0.0630	0.0040	0.9503
	0.9	0.2012	0.0012	0.0882	0.0078	0.9470	0.1998	-0.0002	0.0649	0.0042	0.9508

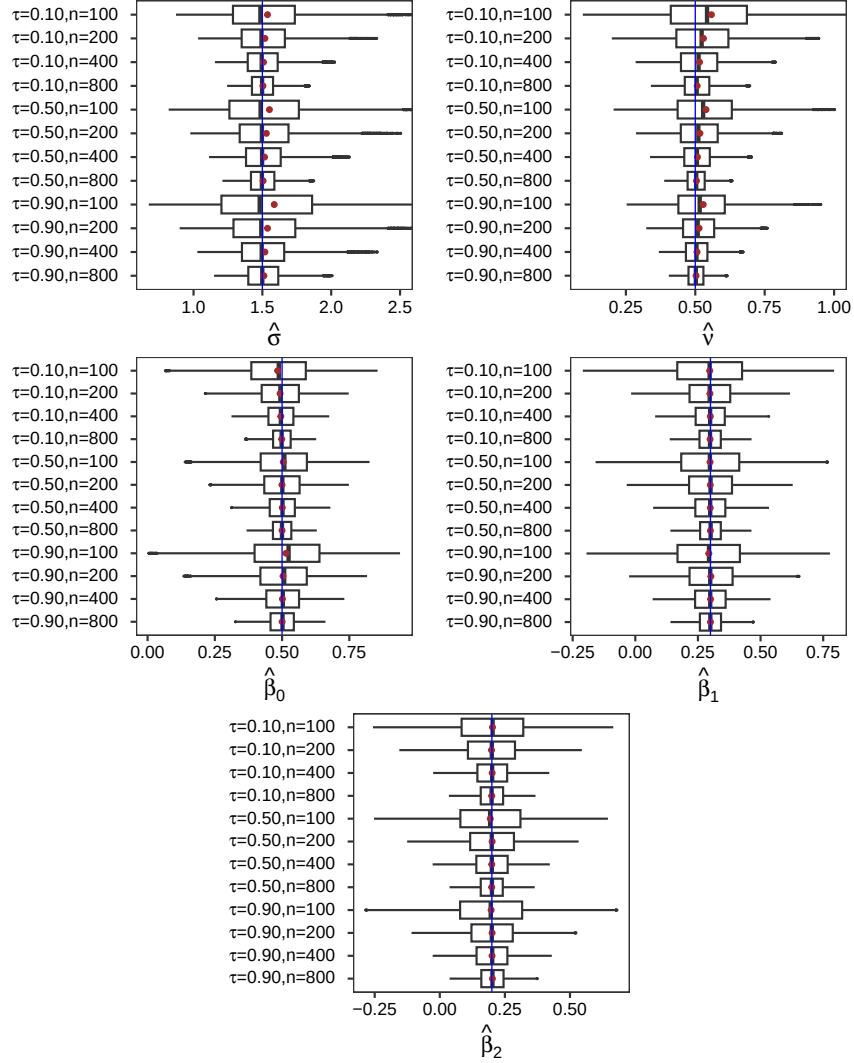


FIGURE 11: Boxplots of the maximum likelihood estimates obtained from Monte Carlo simulations for the NGG regression model with $\sigma = 1.5$, $\nu = 0.5$, $\beta_0 = 0.5$, $\beta_1 = 0.3$ and $\beta_2 = 0.2$, considering $n \in \{100, 200, 400, 800\}$ and $\tau \in \{0.10, 0.50, 0.90\}$. Legend: —: true parameter value; •: mean of the 10,000 replications.

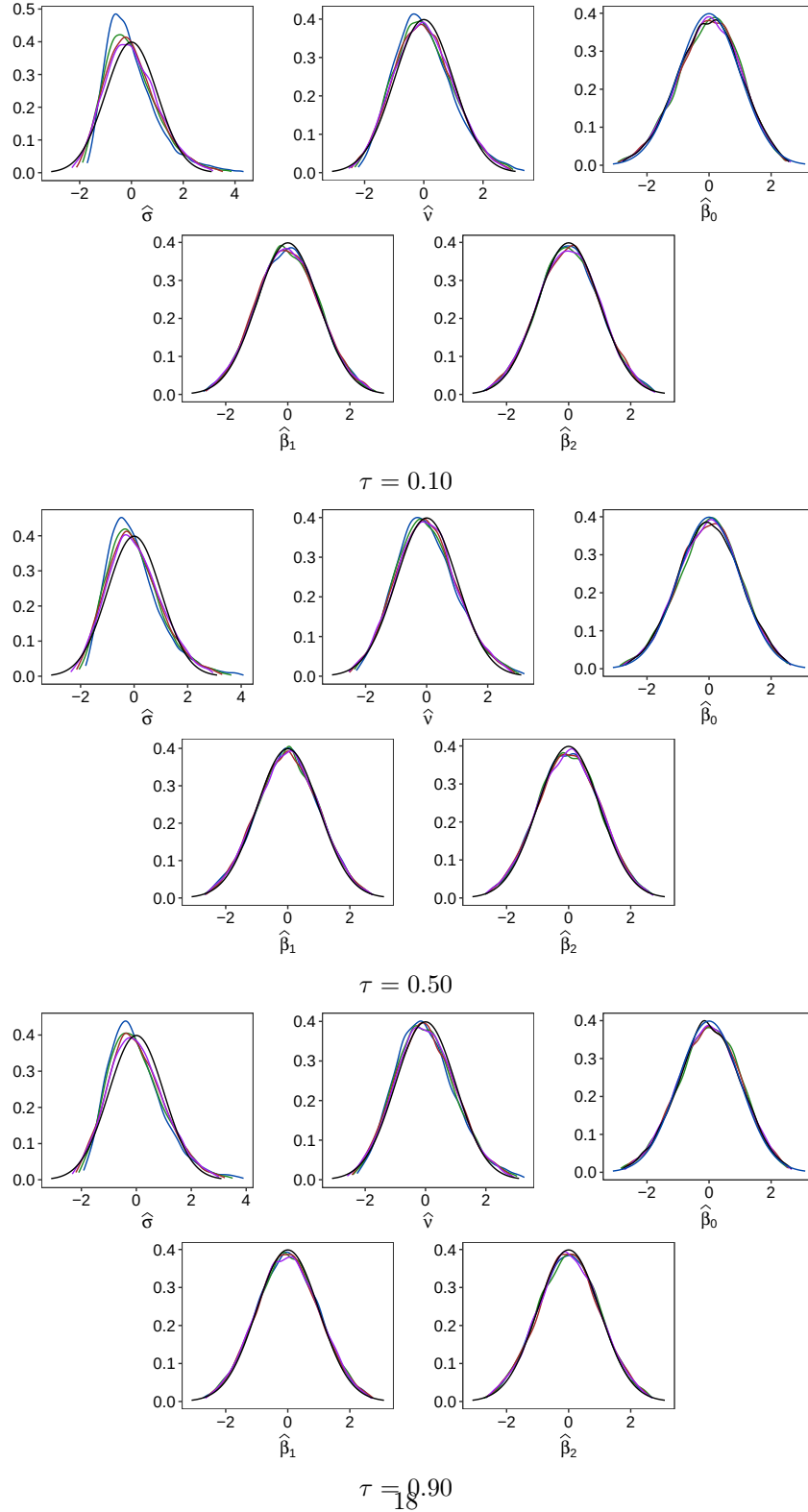


FIGURE 12: Sampling densities of the maximum likelihood estimators of the parameters of the NGG regression model obtained via Monte Carlo simulation, considering $\sigma = 1.5$, $\nu = 0.5$, $\beta = (0.5, 0.3, 0.2)$, $\tau \in \{0.10, 0.50, 0.90\}$ and $n \in \{100, 200, 400, 800\}$. Legend: $n = 100$; $n = 200$; $n = 400$; $n = 800$; $N(0,1)$.

4.2. Scenario 3 – $\sigma = 1.5$, $\nu = 2$, $\beta = (0.3, 0.5, 0.2)$

TABLE 6: Monte Carlo simulation results for the maximum likelihood estimators of the NGG regression model with $\sigma = 1.5$, $\nu = 2.0$, $\beta_0 = 0.3$, $\beta_1 = 0.5$, $\beta_2 = 0.2$, $n \in \{100, 200, 400, 800\}$, and $\tau \in \{0.10, 0.50, 0.90\}$.

par	τ	$n = 100$					$n = 200$				
		mean	bias	se	mse	cp	mean	bias	se	mse	cp
$\hat{\sigma}$	0.1	1.6790	0.1790	1.0311	1.0952	0.8811	1.5833	0.0833	0.5822	0.3459	0.9252
	0.5	1.6598	0.1598	0.8550	0.7566	0.9169	1.5668	0.0668	0.5029	0.2574	0.9364
	0.9	1.6205	0.1205	0.6938	0.4959	0.9299	1.5479	0.0479	0.4067	0.1677	0.9424
$\hat{\nu}$	0.1	2.2390	0.2390	0.7647	0.6419	0.9551	2.0863	0.0863	0.4111	0.1764	0.9513
	0.5	2.1627	0.1627	0.6419	0.4385	0.9550	2.0720	0.0720	0.3718	0.1434	0.9544
	0.9	2.1293	0.1293	0.5587	0.3289	0.9586	2.0630	0.0630	0.3410	0.1203	0.9611
$\hat{\beta}_0$	0.1	0.3060	0.0060	0.0841	0.0071	0.9349	0.3043	0.0043	0.0578	0.0034	0.9388
	0.5	0.3025	0.0025	0.0656	0.0043	0.9417	0.3003	0.0003	0.0451	0.0020	0.9474
	0.9	0.2925	-0.0075	0.0625	0.0040	0.9417	0.2962	-0.0038	0.0431	0.0019	0.9479
$\hat{\beta}_1$	0.1	0.4968	-0.0032	0.0750	0.0056	0.9352	0.4981	-0.0019	0.0550	0.0030	0.9397
	0.5	0.4961	-0.0039	0.0793	0.0063	0.9409	0.4990	-0.0010	0.0537	0.0029	0.9467
	0.9	0.5006	0.0006	0.0739	0.0055	0.9415	0.5005	0.0005	0.0522	0.0027	0.9476
$\hat{\beta}_2$	0.1	0.1989	-0.0011	0.0762	0.0058	0.9333	0.1979	-0.0021	0.0511	0.0026	0.9389
	0.5	0.1985	-0.0015	0.0775	0.0060	0.9456	0.2003	0.0003	0.0515	0.0027	0.9470
	0.9	0.2018	0.0018	0.0733	0.0054	0.9462	0.2000	0.0000	0.0495	0.0024	0.9480
par	τ	$n = 400$					$n = 800$				
		mean	bias	se	mse	cp	mean	bias	se	mse	cp
$\hat{\sigma}$	0.1	1.5365	0.0365	0.3674	0.1363	0.9360	1.5177	0.0177	0.2458	0.0607	0.9428
	0.5	1.5299	0.0299	0.3170	0.1014	0.9412	1.5122	0.0122	0.2130	0.0455	0.9452
	0.9	1.5213	0.0213	0.2692	0.0729	0.9491	1.5138	0.0138	0.1847	0.0343	0.9525
$\hat{\nu}$	0.1	2.0405	0.0405	0.2665	0.0727	0.9514	2.0185	0.0185	0.1813	0.0332	0.9508
	0.5	2.0330	0.0330	0.2468	0.0620	0.9493	2.0172	0.0172	0.1681	0.0286	0.9532
	0.9	2.0308	0.0308	0.2304	0.0540	0.9550	2.0115	0.0115	0.1599	0.0257	0.9513
$\hat{\beta}_0$	0.1	0.3018	0.0018	0.0418	0.0018	0.9440	0.3008	0.0008	0.0292	0.0009	0.9475
	0.5	0.2999	-0.0001	0.0321	0.0010	0.9514	0.3001	0.0001	0.0217	0.0005	0.9515
	0.9	0.2982	-0.0018	0.0310	0.0010	0.9528	0.2992	-0.0008	0.0218	0.0005	0.9514
$\hat{\beta}_1$	0.1	0.4986	-0.0014	0.0388	0.0015	0.9475	0.4997	-0.0003	0.0277	0.0008	0.9452
	0.5	0.4999	-0.0001	0.0391	0.0015	0.9480	0.4999	-0.0001	0.0257	0.0007	0.9544
	0.9	0.5001	0.0001	0.0359	0.0013	0.9508	0.4998	-0.0002	0.0252	0.0006	0.9532
$\hat{\beta}_2$	0.1	0.1998	-0.0002	0.0373	0.0014	0.9433	0.1996	-0.0004	0.0251	0.0006	0.9502
	0.5	0.2000	0.0000	0.0360	0.0013	0.9512	0.1999	-0.0001	0.0262	0.0007	0.9506
	0.9	0.2002	0.0002	0.0377	0.0014	0.9536	0.2000	0.0000	0.0259	0.0007	0.9511

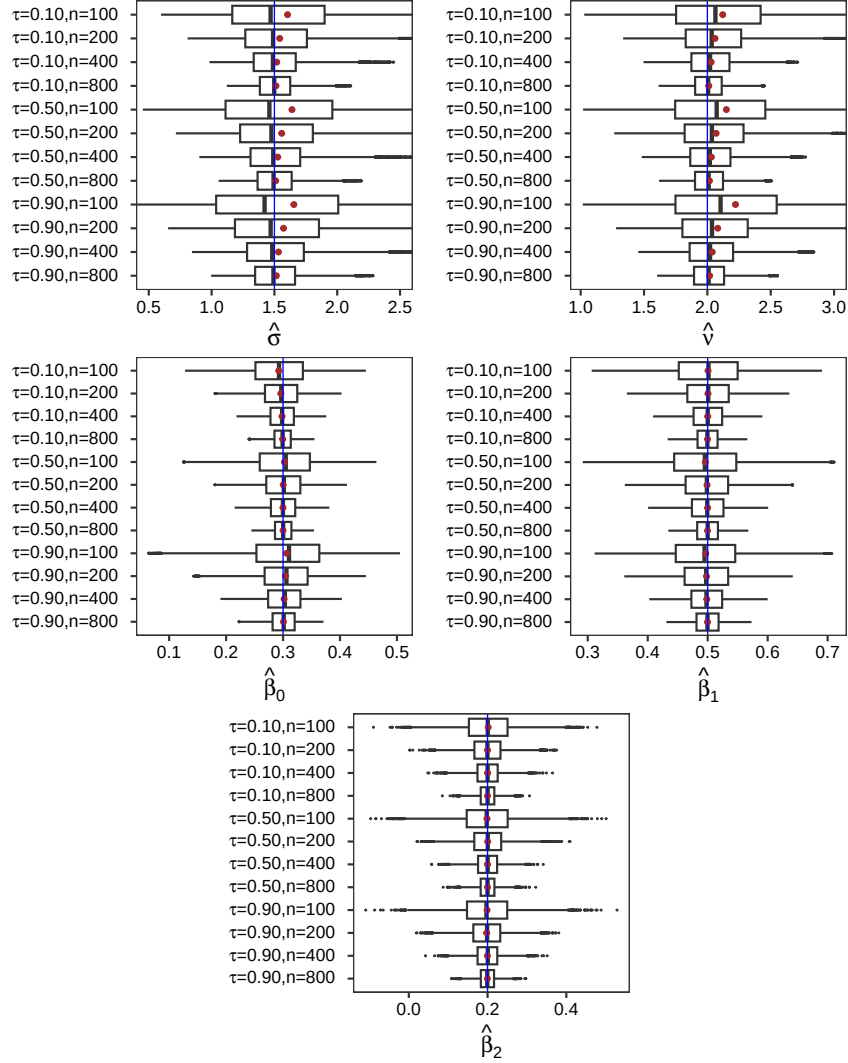


FIGURE 13: Boxplots of the maximum likelihood estimates obtained from Monte Carlo simulations for the NGG regression model with $\sigma = 1.5$, $\nu = 2$, $\beta_0 = 0.3$, $\beta_1 = 0.5$ and $\beta_2 = 0.2$, considering $n \in \{100, 200, 400, 800\}$ and $\tau \in \{0.10, 0.50, 0.90\}$. Legend: —: true parameter value; •: mean of the 10,000 replications.

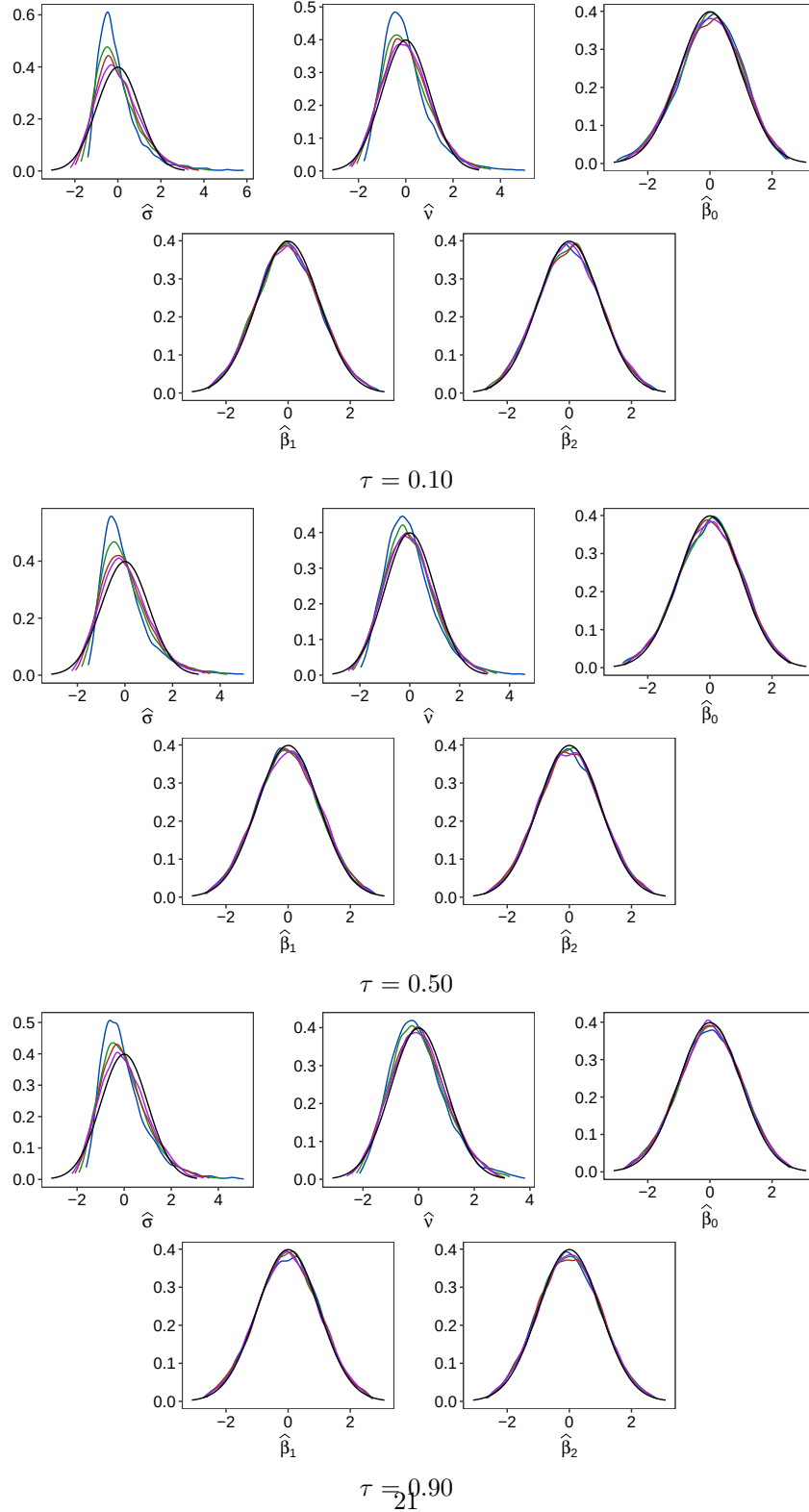


FIGURE 14: Sampling densities of the maximum likelihood estimators of the parameters of the NGGD(μ_τ, σ, ν) distribution obtained via Monte Carlo simulation, considering $\sigma = 1.5$, $\nu = 2$, $\beta_0 = 0.3$, $\beta_1 = 0.5$ and $\beta_2 = 0.2$ and $n \in \{100, 200, 400, 800\}$. Legend: $n = 100$; $n = 200$; $n = 400$; $n = 800$; $N(0,1)$.

4.3. Scenario 6 – $\sigma = 1.0$, $\nu = 0.4$, $\beta = (1.1, -0.9, 2.0)$

TABLE 7: Monte Carlo simulation results for the maximum likelihood estimators of the NGG regression model with $\sigma = 1.0$, $\nu = 0.4$, $\beta_0 = 1.1$, $\beta_1 = -0.9$, $\beta_2 = 2.0$, $n \in \{100, 200, 400, 800\}$, and $\tau \in \{0.10, 0.50, 0.90\}$.

par	τ	$n = 100$					$n = 200$				
		mean	bias	se	mse	cp	mean	bias	se	mse	cp
$\hat{\sigma}$	0.1	1.0574	0.0574	0.3947	0.1591	0.9264	1.0230	0.0230	0.2294	0.0532	0.9407
	0.5	1.0269	0.0269	0.2704	0.0739	0.9457	1.0153	0.0153	0.1866	0.0350	0.9483
	0.9	1.0260	0.0260	0.2246	0.0511	0.9497	1.0115	0.0115	0.1523	0.0233	0.9512
$\hat{\nu}$	0.1	0.4267	0.0267	0.1059	0.0119	0.9568	0.4115	0.0115	0.0608	0.0038	0.9527
	0.5	0.4268	0.0268	0.0926	0.0093	0.9590	0.4120	0.0120	0.0597	0.0037	0.9530
	0.9	0.4278	0.0278	0.0932	0.0095	0.9634	0.4135	0.0135	0.0614	0.0040	0.9569
$\hat{\beta}_0$	0.1	1.1137	0.0137	0.1880	0.0355	0.9293	1.1095	0.0095	0.1250	0.0157	0.9405
	0.5	1.1078	0.0078	0.1148	0.0132	0.9391	1.1025	0.0025	0.0842	0.0071	0.9449
	0.9	1.0864	-0.0136	0.1233	0.0154	0.9439	1.0950	-0.0050	0.0795	0.0063	0.9480
$\hat{\beta}_1$	0.1	-0.8919	0.0081	0.1395	0.0195	0.9349	-0.8964	0.0036	0.0920	0.0085	0.9436
	0.5	-0.8928	0.0072	0.1205	0.0146	0.9411	-0.8974	0.0026	0.0810	0.0066	0.9454
	0.9	-0.8943	0.0057	0.1373	0.0189	0.9432	-0.9004	-0.0004	0.0797	0.0064	0.9484
$\hat{\beta}_2$	0.1	1.9822	-0.0178	0.2185	0.0481	0.9302	1.9881	-0.0119	0.1323	0.0176	0.9417
	0.5	1.9811	-0.0189	0.1333	0.0181	0.9404	1.9945	-0.0055	0.0970	0.0094	0.9455
	0.9	1.9988	-0.0012	0.1464	0.0214	0.9432	1.9983	-0.0017	0.0980	0.0096	0.9472
par	τ	$n = 400$					$n = 800$				
		mean	bias	se	mse	cp	mean	bias	se	mse	cp
$\hat{\sigma}$	0.1	1.0142	0.0142	0.1553	0.0243	0.9466	1.0058	0.0058	0.1055	0.0112	0.9456
	0.5	1.0080	0.0080	0.1233	0.0153	0.9491	1.0032	0.0032	0.0863	0.0075	0.9514
	0.9	1.0052	0.0052	0.1040	0.0108	0.9494	1.0029	0.0029	0.0717	0.0051	0.9538
$\hat{\nu}$	0.1	0.4050	0.0050	0.0410	0.0017	0.9511	0.4026	0.0026	0.0283	0.0008	0.9455
	0.5	0.4051	0.0051	0.0406	0.0017	0.9497	0.4030	0.0030	0.0280	0.0008	0.9528
	0.9	0.4067	0.0067	0.0420	0.0018	0.9492	0.4029	0.0029	0.0291	0.0009	0.9495
$\hat{\beta}_0$	0.1	1.1061	0.0061	0.0874	0.0077	0.9436	1.1013	0.0013	0.0641	0.0041	0.9448
	0.5	1.1020	0.0020	0.0589	0.0035	0.9517	1.1003	0.0003	0.0418	0.0018	0.9475
	0.9	1.0976	-0.0024	0.0590	0.0035	0.9499	1.0989	-0.0011	0.0393	0.0015	0.9526
$\hat{\beta}_1$	0.1	-0.8982	0.0018	0.0644	0.0041	0.9486	-0.8986	0.0014	0.0459	0.0021	0.9503
	0.5	-0.8990	0.0010	0.0583	0.0034	0.9495	-0.8998	0.0002	0.0412	0.0017	0.9518
	0.9	-0.8999	0.0001	0.0581	0.0034	0.9481	-0.9002	-0.0002	0.0407	0.0017	0.9529
$\hat{\beta}_2$	0.1	1.9935	-0.0065	0.0937	0.0088	0.9448	1.9983	-0.0017	0.0676	0.0046	0.9483
	0.5	1.9965	-0.0035	0.0667	0.0045	0.9521	1.9990	-0.0010	0.0471	0.0022	0.9493
	0.9	1.9996	-0.0004	0.0661	0.0044	0.9492	1.9998	-0.0002	0.0455	0.0021	0.9553

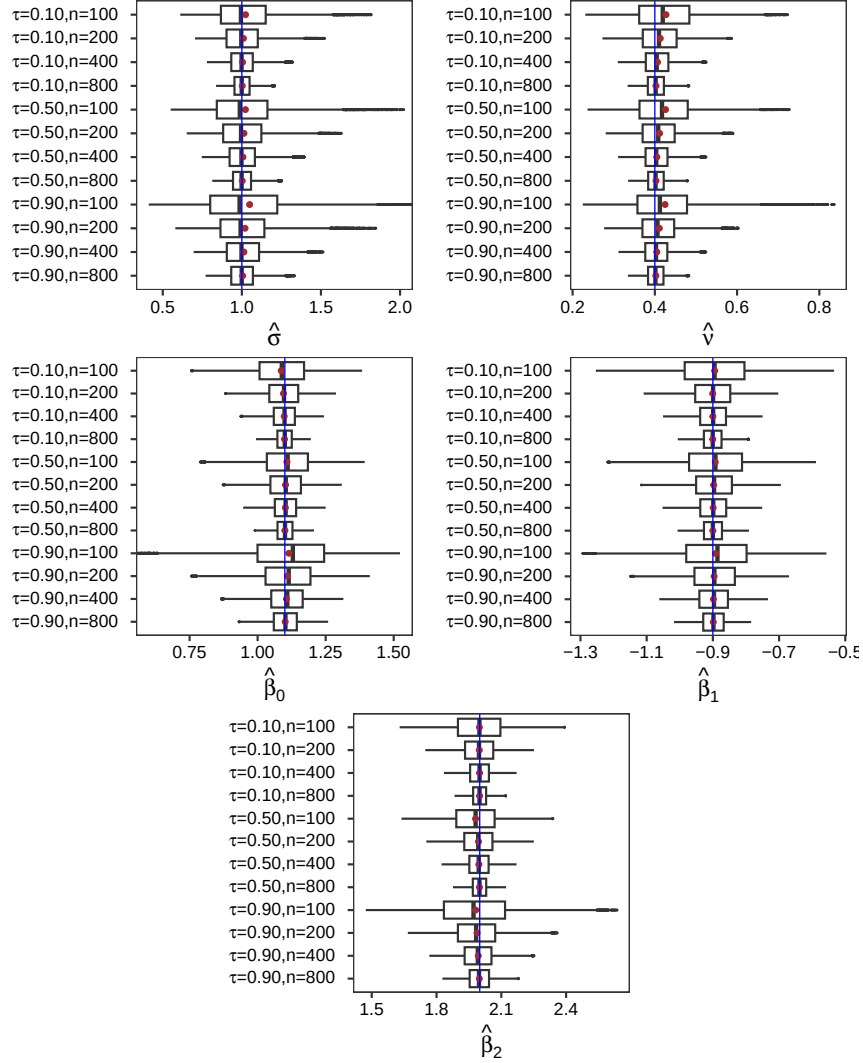


FIGURE 15: Boxplots of the maximum likelihood estimates obtained from Monte Carlo simulations for the NGG regression model with $\sigma = 1.0$, $\nu = 0.4$, $\beta_0 = 1.1$, $\beta_1 = -0.9$ and $\beta_2 = 2.0$, considering $n \in \{100, 200, 400, 800\}$ and $\tau \in \{0.10, 0.50, 0.90\}$. Legend: —: true parameter value; •: mean of the 10,000 replications.

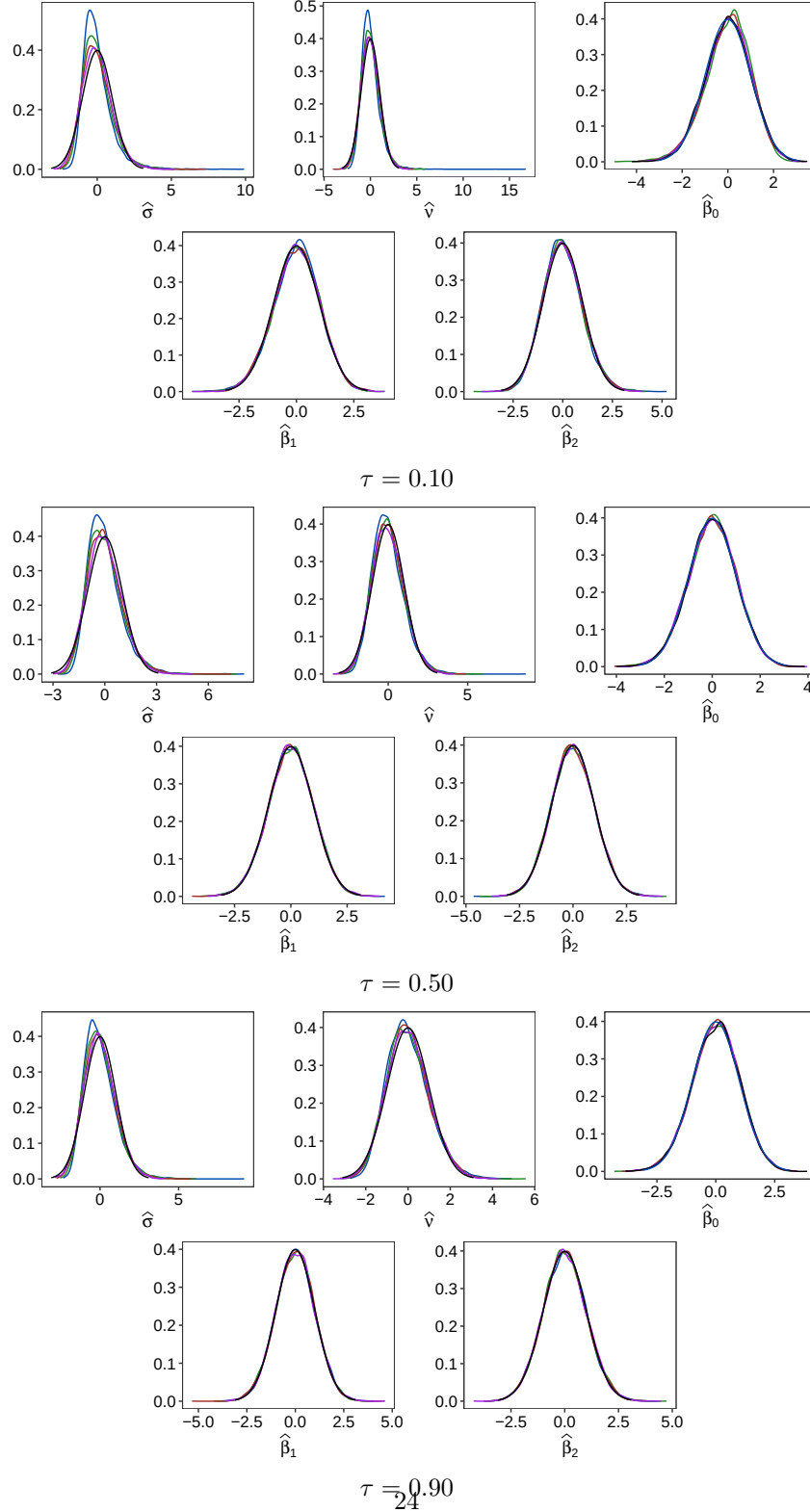


FIGURE 16: Sampling densities of the maximum likelihood estimators of the parameters of the NGG regression model obtained via Monte Carlo simulation, considering $\sigma = 1.0$, $\nu = 0.4$, $\beta_0 = 1.1$, $\beta_1 = -0.9$ and $\beta_2 = 2.0$ and $n \in \{100, 200, 400, 800\}$. Legend: $n = 100$; $n = 200$; $n = 400$; $n = 800$; $N(0,1)$.

5. Application

In this section, we present the results of applying the proposed distribution to the datasets analyzed, considering all quantile levels adopted in the study, namely, $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$.

TABLE 8: Results of fitting the QGG, EBXII_I, EBXII_II, EL, EDW, and TD distributions to the datasets analyzed: parameter estimates $\hat{\mu}_\tau$, $\hat{\sigma}$, and $\hat{\nu}$ (standard errors in parentheses) and goodness-of-fit statistics (W, A, KS, AIC, BIC, HQIC), with $\tau = 0.10$.

Data	Dist.	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	W	A	KS	AIC	BIC	HQIC
Social media	QGG	1.0634 (0.0509)	1.5258 (0.1167)	0.3917 (0.0314)	0.1026	0.7228	0.0302	3336.6700	3351.2740	3342.2310
	EBXII_I	0.9580 (0.0279)	4.4183 (0.4655)	1.3386 (0.0855)	7.6111	44.0157	0.1616	3915.2220	3929.8260	3920.7840
	EBXII_II	0.9648 (0.0281)	1.8264 (0.1852)	1.0784 (0.0851)	7.5798	43.8454	0.1558	3913.4310	3928.0350	3918.9920
	EL	1.0117 (0.0338)	4.2535 (0.2975)	0.2072 (0.0210)	4.2595	25.5078	0.1166	3679.7570	3694.3610	3685.3180
	EDW	1.0646 (0.0495)	4.0934 (0.1447)	3.2699 (0.2627)	0.1331	0.9151	0.0343	3337.9820	3352.5860	3343.5440
	TD	1.0808 (0.0358)	2.4592 (0.0718)	8.5829 (1.0654)	3.8782	23.3712	0.1134	3635.3370	3649.9410	3640.8990
Life choices	QGG	0.6808 (0.0217)	3.0818 (2.8227)	4.0152 (1.9646)	0.1159	0.9329	0.0623	-119.6953	-110.7651	-116.0666
	EBXII_I	0.6580 (0.0176)	4.0726 (1.6556)	3.3180 (0.6416)	0.6972	4.5291	0.1190	-69.2868	-60.3566	-65.6582
	EBXII_II	0.6712 (0.0181)	2.7504 (0.4449)	5.4556 (1.0287)	0.4228	2.9093	0.0985	-90.8875	-81.9573	-87.2588
	EL	0.6106 (0.0170)	6.3839 (1.4561)	1.5696 (0.7394)	1.4294	8.5637	0.2142	-7.0554	1.8748	-3.4267
	EDW	0.6621 (0.0186)	0.7245 (0.0796)	2.9219 (0.5736)	0.4120	2.8508	0.0956	-92.3933	-83.4631	-88.7647
	TD	0.6620 (0.0208)	7.8515 (0.8916)	0.4989 (0.1358)	0.4458	3.0407	0.1406	-82.2735	-73.3433	-78.6449
Grade point average	QGG	3.1296 (0.0192)	2.7282 (0.6380)	1.8394 (0.2269)	5.2590	35.5750	0.1583	904.8910	919.5750	910.4755
	EBXII_I	2.0589 (0.0291)	10.1806 (0.5496)	4.6366 (0.4739)	8.3708	50.9169	0.3941	3116.7090	3131.3930	3122.2940
	EBXII_II	2.9569 (0.0149)	4.4519 (0.7454)	1.6861 (0.2263)	10.9790	63.9774	0.2644	1609.2490	1623.9330	1614.8340
	EL	2.9594 (0.0158)	8.3760 (1.2618)	1.0181 (0.6505)	10.7849	63.0106	0.2764	1604.4000	1619.0840	1609.9840
	EDW	3.0788 (0.0184)	3.2436 (0.1187)	5.2387 (0.6038)	6.0829	39.4577	0.1706	1029.8560	1044.5400	1035.4400
	TD	2.9488 (0.0147)	6.5371 (0.1384)	0.7323 (25.7631)	11.4674	66.4006	0.2694	1658.3950	1673.0790	1663.9790

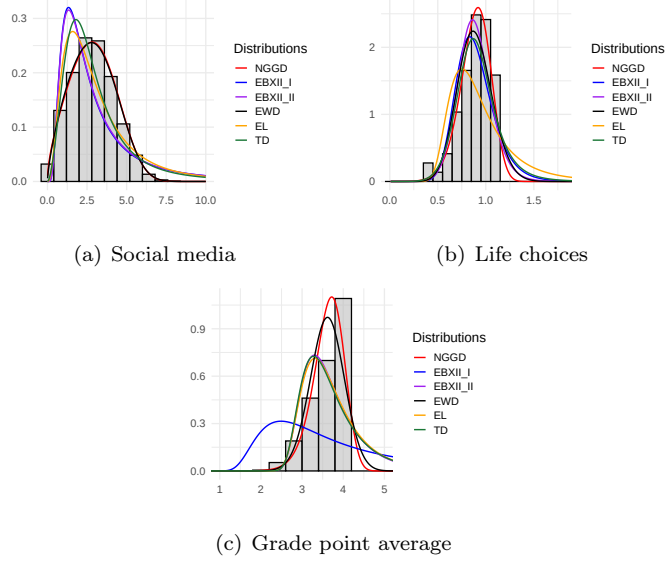


FIGURE 17: Histograms and fitted density curves of the QGG, EBXII_I, EBXII_II, EL, EWD, and TD distributions for the three analyzed datasets, $\tau = 0.10$.

TABLE 9: Results of fitting the QGG, EBXII_I, EBXII_II, EL, EDW, and TD distributions to the datasets analyzed: parameter estimates $\hat{\mu}_\tau$, $\hat{\sigma}$, and $\hat{\nu}$ (standard errors in parentheses) and goodness-of-fit statistics (W, A, KS, AIC, BIC, HQIC), with $\tau = 0.25$.

Data	Dist.	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	W	A	KS	AIC	BIC	HQIC
Social media	QGG	1.8542 (0.0521)	1.5292 (0.1169)	0.3909 (0.0314)	0.1031	0.7263	0.0301	3336.6690	3351.2730	3342.2300
	EBXII_I	1.4475 (0.0355)	5.0735 (0.6321)	1.2456 (0.0890)	7.5850	43.8745	0.1601	3912.4520	3927.0560	3918.0140
	EBXII_II	1.4525 (0.0357)	1.6232 (0.1623)	1.1800 (0.0898)	7.5770	43.8310	0.1581	3911.9110	3926.5150	3917.4720
	EL	1.5811 (0.0410)	4.2628 (0.2985)	0.2064 (0.0209)	4.2492	25.4495	0.1159	3679.3960	3694.0000	3684.9570
	EDW	1.8377 (0.0503)	4.0934 (0.1447)	3.2699 (0.2627)	0.1331	0.9151	0.0343	3337.9820	3352.5860	3343.5440
	TD	1.6675 (0.0393)	2.5546 (0.0742)	10.4109 (1.3175)	3.6169	21.8818	0.1075	3615.0180	3629.6220	3620.5790
Life choices	QGG	0.7860 (0.0170)	4.1376 (2.5652)	3.3818 (1.1370)	0.1655	1.2763	0.0620	-114.6942	-105.7640	-111.0655
	EBXII_I	0.7623 (0.0159)	1.5698 (0.4358)	5.5013 (0.8886)	0.4143	2.8580	0.1040	-91.2094	-82.2792	-87.5808
	EBXII_II	0.7567 (0.0160)	3.2249 (0.5108)	4.3694 (0.8734)	0.5383	3.6015	0.1179	-81.6217	-72.6915	-77.9931
	EL	0.7105 (0.0211)	5.1887 (2.3394)	2.5375 (3.1361)	1.5387	9.1414	0.2234	0.8903	9.8205	4.5189
	EDW	0.7549 (0.0167)	0.6735 (0.0816)	2.5663 (0.4691)	0.4768	3.2428	0.1104	-86.4338	-77.5035	-82.8051
	TD	0.7844 (0.0175)	16.8130 (2.4870)	1.1900 (0.3688)	0.0901	0.7653	0.0799	-118.6139	-109.6837	-114.9852
Grade point average	QGG	3.3998 (0.0243)	0.9327 (0.7981)	3.4020 (1.9378)	4.7780	33.7276	0.1709	784.7110	799.3951	790.2955
	EBXII_I	2.7250 (0.0333)	17.4482 (0.9893)	6.2053 (1.2164)	8.5708	51.9215	0.3968	2815.9650	2830.6490	2821.5490
	EBXII_II	3.1987 (0.0156)	4.5151 (0.7879)	1.6684 (0.2330)	10.9861	64.0128	0.2652	1608.8470	1623.5310	1614.4320
	EL	3.2034 (0.0163)	9.4833 (0.7908)	0.6526 (0.1648)	10.5689	61.9361	0.2719	1584.0460	1598.7300	1589.6300
	EDW	3.3307 (0.0156)	3.1781 (0.1025)	4.8686 (0.4496)	6.2077	40.0742	0.1795	1047.8810	1062.5650	1053.4660
	TD	3.1871 (0.0164)	6.5431 (0.1384)	0.7474 (25.8970)	11.4734	66.4304	0.2692	1658.3840	1673.0680	1663.9690

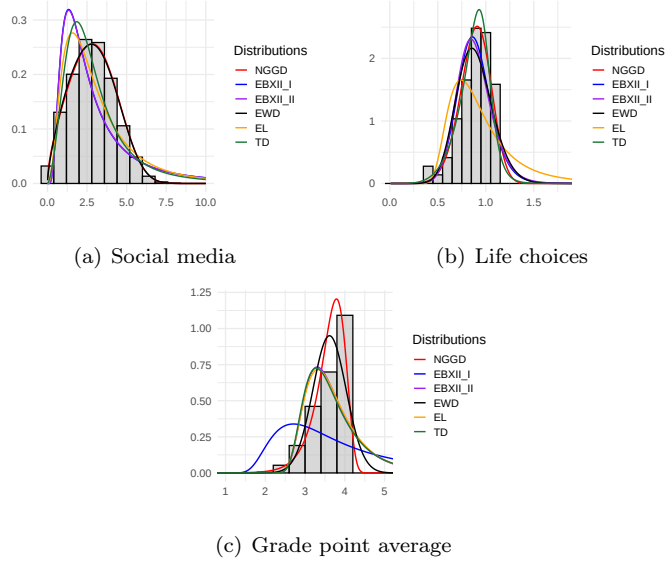


FIGURE 18: Histograms and fitted density curves of the QGG, EBXII_I, EBXII_II, EL, EWD, and TD distributions for the three analyzed datasets, $\tau = 0.25$.

TABLE 10: Results of fitting the QGG, EBXII_I, EBXII_II, EL, EDW, and TD distributions to the datasets analyzed: parameter estimates $\hat{\mu}_\tau$, $\hat{\sigma}$, and $\hat{\nu}$ (standard errors in parentheses) and goodness-of-fit statistics (W, A, KS, AIC, BIC, HQIC), with $\tau = 0.75$.

Data	Dist.	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	W	A	KS	AIC	BIC	HQIC
Social media	QGG	3.9190 (0.0558)	1.5293 (0.1169)	0.3909 (0.0314)	0.1031	0.7264	0.0301	3336.6690	3351.2730	3342.2300
	EBXII_I	4.1355 (0.1309)	5.4216 (0.7157)	1.2025 (0.0888)	7.5769	43.8308	0.1588	3911.9670	3926.5710	3917.5290
	EBXII_II	4.0243 (0.1240)	3.2392 (0.4279)	0.6898 (0.0801)	8.1129	46.7064	0.1815	3961.1940	3975.7980	3966.7560
	EL	4.1103 (0.1024)	4.3497 (0.2902)	0.2028 (0.0195)	4.2254	25.3140	0.1171	3676.0500	3690.6540	3681.6110
	EDW	3.9072 (0.0572)	4.0921 (0.1447)	3.2671 (0.2623)	0.1333	0.9164	0.0343	3337.9830	3352.5860	3343.5440
	TD	3.9286 (0.1023)	2.3712 (0.0665)	6.3424 (0.6748)	4.4512	26.6148	0.1128	3670.2270	3684.8310	3675.7880
Life choices	QGG	1.0015 (0.0134)	3.0705 (1.3262)	3.9404 (0.9366)	0.1188	0.9534	0.0578	-119.3970	-110.4668	-115.7684
	EBXII_I	0.9931 (0.0163)	1.3823 (0.4664)	6.1492 (1.2331)	0.3659	2.5611	0.0917	-95.4688	-86.5386	-91.8401
	EBXII_II	0.9992 (0.0192)	3.2215 (0.5055)	4.2955 (0.8496)	0.5488	3.6637	0.1106	-80.9805	-72.0503	-77.3519
	EL	1.0766 (0.0315)	9.5681 (1.9448)	0.7848 (0.2450)	1.2966	7.8546	0.2023	-18.3031	-9.3729	-14.6745
	EDW	1.0237 (0.0260)	0.4269 (0.1682)	1.5675 (0.4967)	0.7804	5.0120	0.1350	-62.3449	-53.4147	-58.7163
	TD	1.0201 (0.0189)	8.8876 (1.0433)	0.5850 (0.1653)	0.3535	2.4815	0.1197	-91.1964	-82.2662	-87.5678
Grade point average	QGG	3.8994 (0.0072)	0.2973 (0.0367)	8.3702 (0.8844)	4.6432	34.0682	0.1888	628.8811	643.5651	634.4656
	EBXII_I	4.7320 (0.0655)	42.0955 (2.6090)	14.0325 (96.8633)	8.9218	53.6839	0.3902	2421.0690	2435.7530	2426.6540
	EBXII_II	4.0384 (0.0291)	4.4284 (0.5533)	1.6909 (0.1695)	10.9680	63.9224	0.2644	1609.4290	1624.1130	1615.0140
	EL	4.0471 (0.0300)	6.9358 (0.1447)	5.4750 (0.5437)	11.3505	65.8209	0.2672	1645.4400	1660.1240	1651.0240
	EDW	3.9206 (0.0196)	1.9057 (0.1360)	2.1522 (0.1460)	7.8654	48.3884	0.2082	1260.0070	1274.6910	1265.5910
	TD	4.0531 (0.0332)	6.5429 (0.1385)	0.7455 (25.8368)	11.4733	66.4300	0.2692	1658.3850	1673.0690	1663.9700

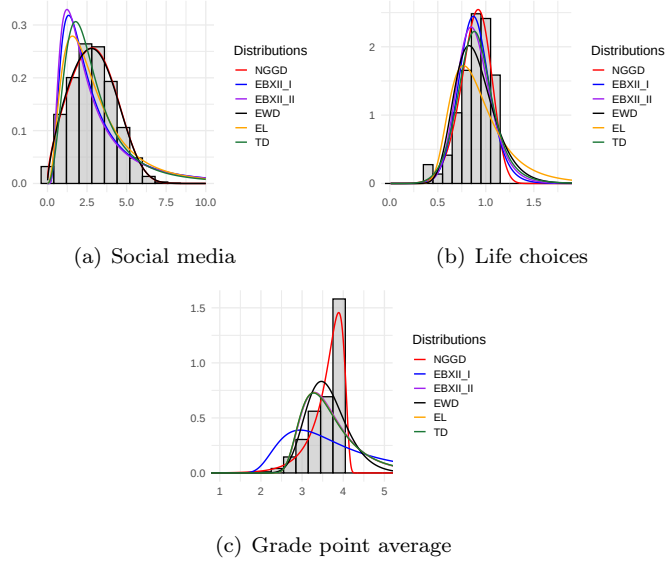


FIGURE 19: Histograms and fitted density curves of the QGG, EBXII_I, EBXII_II, EL, EWD, and TD distributions for the three analyzed datasets, $\tau = 0.75$.

TABLE 11: Results of fitting the QGG, EBXII_I, EBXII_II, EL, EDW, and TD distributions to the datasets analyzed: parameter estimates $\hat{\mu}_\tau$, $\hat{\sigma}$, and $\hat{\nu}$ (standard errors in parentheses) and goodness-of-fit statistics (W, A, KS, AIC, BIC, HQIC), with $\tau = 0.90$.

Data	Dist.	$\hat{\mu}_\tau$	$\hat{\sigma}$	$\hat{\nu}$	W	A	KS	AIC	BIC	HQIC
Social media	QGG	4.8172 (0.0634)	1.5293 (0.1169)	0.3909 (0.0314)	0.1031	0.7264	0.0301	3336.6690	3351.2730	3342.2300
	EBXII_I	7.4500 (0.3147)	5.1693 (0.6417)	1.2308 (0.0870)	7.5833	43.8654	0.1586	3912.1620	3926.7660	3917.7230
	EBXII_II	6.7500 (0.2426)	2.2028 (0.2072)	0.9733 (0.0739)	7.7561	44.7916	0.1854	3926.7790	3941.3830	3932.3400
	EL	6.3477 (0.1949)	5.1087 (0.4331)	0.1604 (0.0184)	3.9248	23.5975	0.1130	3652.0160	3666.6200	3657.5770
	EDW	4.8097 (0.0809)	3.3575 (0.2359)	2.2344 (0.2630)	0.4547	2.9015	0.0569	3360.3690	3374.9730	3365.9310
	TD	6.4206 (0.2357)	2.2249 (0.0648)	4.8150 (0.5349)	4.8490	28.8487	0.1226	3702.6120	3717.2160	3708.1730
Life choices	QGG	1.0929 (0.0176)	3.3169 (1.1040)	3.6451 (0.6901)	0.1365	1.0780	0.0606	-117.2398	-108.3096	-113.6112
	EBXII_I	1.1206 (0.0326)	1.9129 (1.1080)	5.1108 (1.6433)	0.4585	3.1254	0.0922	-88.1961	-79.2659	-84.5674
	EBXII_II	1.1295 (0.0319)	3.3321 (0.5593)	4.1778 (0.9175)	0.5667	3.7692	0.1137	-79.6425	-70.7123	-76.0139
	EL	1.3750 (0.0533)	6.7671 (0.9330)	1.3943 (0.3693)	1.4009	8.4119	0.2118	-8.9676	-0.0374	-5.3389
	EDW	1.1699 (0.0587)	0.4940 (0.2203)	1.7747 (0.7570)	0.7005	4.5556	0.1276	-68.5689	-59.6387	-64.9402
	TD	1.1056 (0.0219)	12.5456 (1.6951)	0.8906 (0.2709)	0.1724	1.3371	0.0761	-109.9677	-101.0375	-106.3390
Grade point average	QGG	4.0451 (0.0140)	1.2917 (0.3761)	2.7707 (0.4954)	4.8907	34.0749	0.1666	822.2500	836.9340	827.8345
	EBXII_I	6.3620 (0.1089)	49.5854 (3.1225)	14.1729 (107.6880)	8.9933	54.0430	0.3906	2358.2450	2372.9290	2363.8300
	EBXII_II	4.6688 (0.0447)	4.5908 (0.5941)	1.6440 (0.1700)	10.9657	63.9113	0.2645	1608.3880	1623.0720	1613.9720
	EL	4.7019 (0.0469)	8.1628 (0.6331)	1.1549 (0.4117)	10.8303	63.2370	0.2713	1609.1600	1623.8440	1614.7440
	EDW	4.4195 (0.0423)	0.6163 (0.1834)	1.1059 (0.1309)	9.2938	55.5711	0.2328	1416.0050	1430.6890	1421.5900
	TD	4.7242 (0.0514)	6.5461 (0.1384)	0.7317 (25.8235)	11.4755	66.4412	0.2689	1658.3920	1673.0760	1663.9770

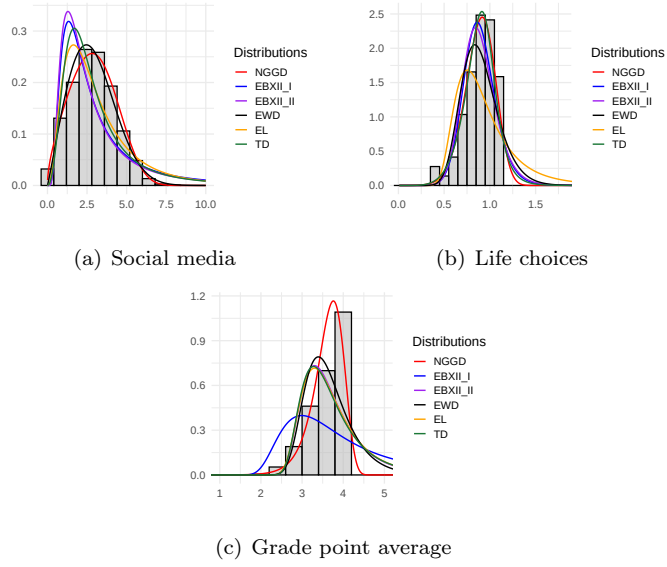


FIGURE 20: Histograms and fitted density curves of the QGG, EBXII_I, EBXII_II, EL, EWD, and TD distributions for the three analyzed datasets, $\tau = 0.90$.